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Foster et al.

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(54) **BULK AND TOTE CART**

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B62B 3/00 (2006.01)

(52) **U.S. Cl.**
CPC ***B62B 3/025*** (2013.01); ***B62B 3/005*** (2013.01)

(58) **Field of Classification Search**
CPC B62B 3/025; B62B 3/005
See application file for complete search history.

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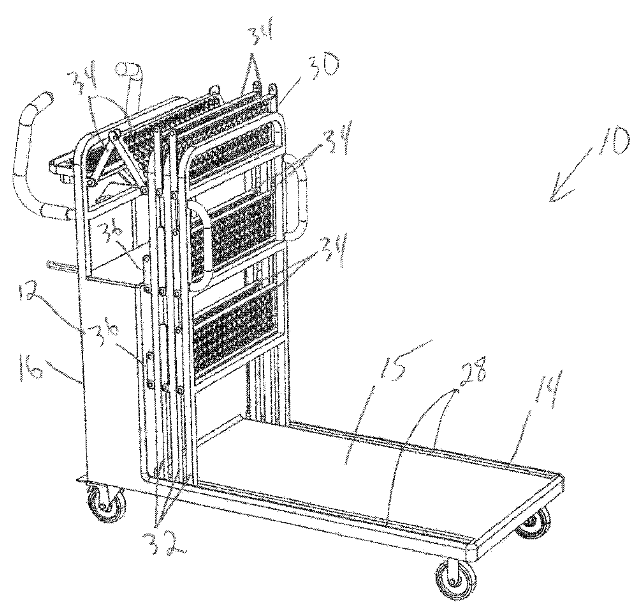
(57) **ABSTRACT**

A cart includes a platform and a plurality of wheels supporting the platform. A plurality of retractable shelves are supported on the platform. The plurality of retractable shelves are reconfigurable between a deployed position and a retracted position. In the deployed position, a plurality of totes (or other smaller objects) can be supported on the shelves. In the retracted position, the cart is in an L Cart configuration. The platform is uncovered by the shelves and a larger item (or several larger items) can be placed on the platform.

20 Claims, 12 Drawing Sheets



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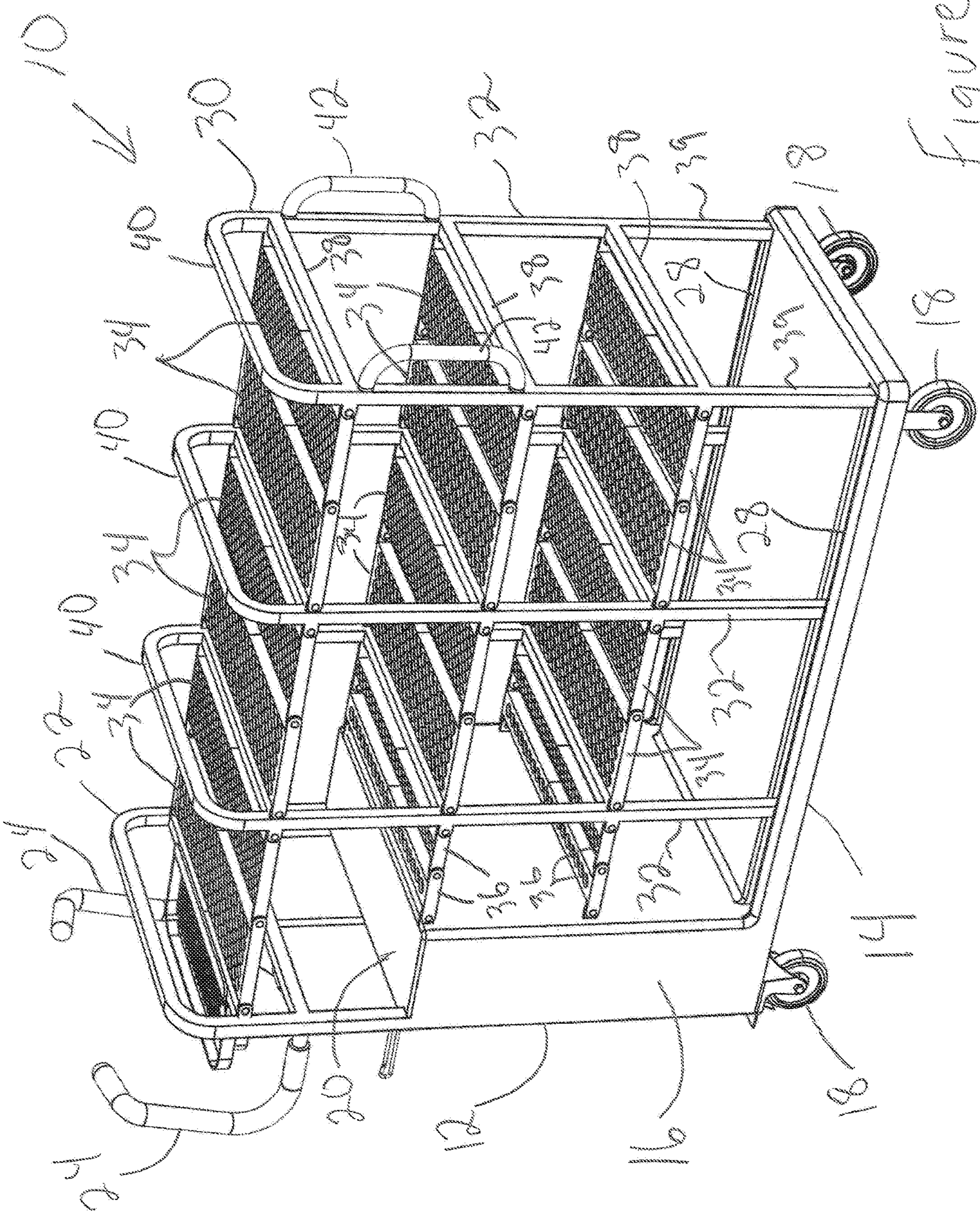

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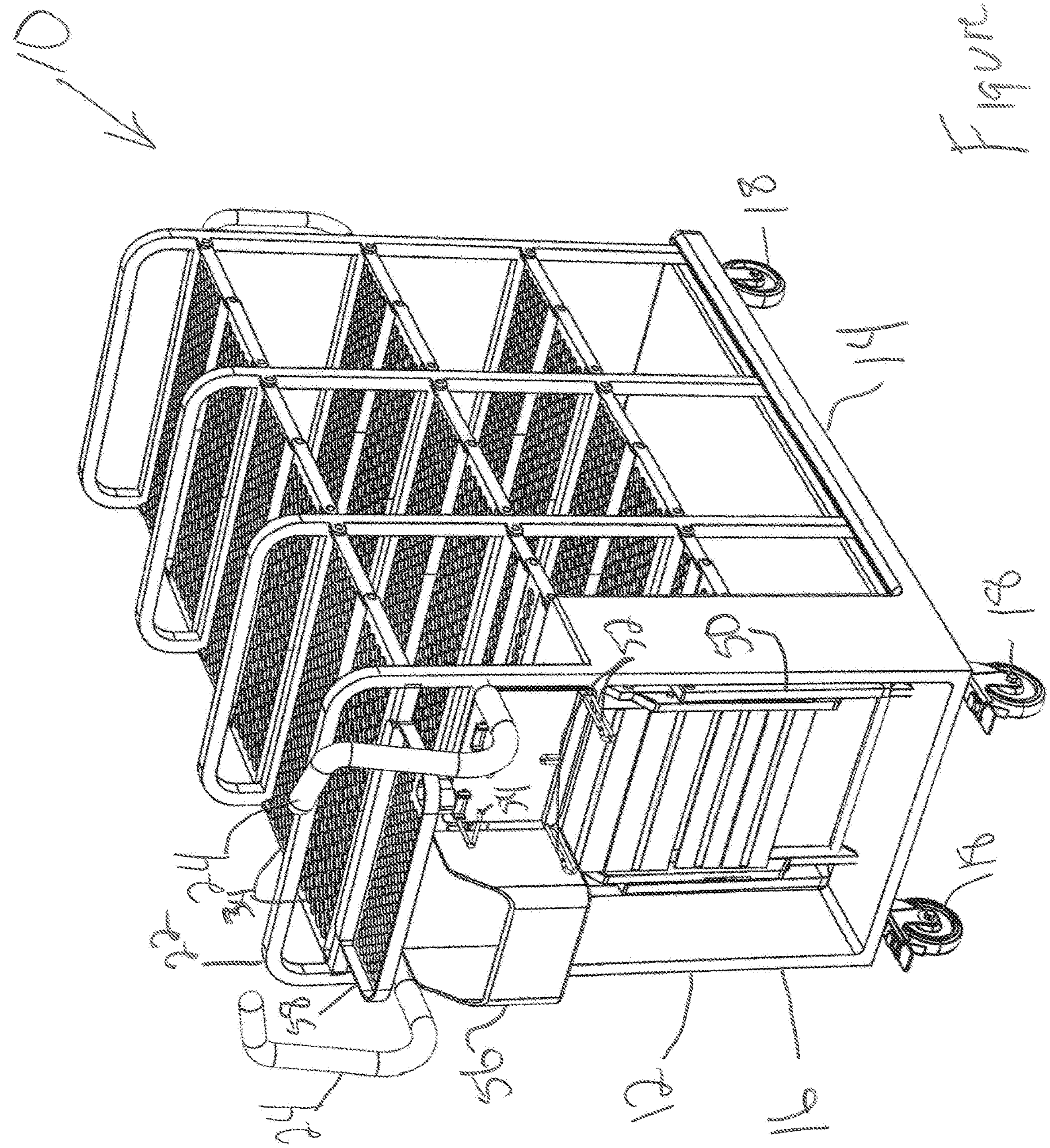
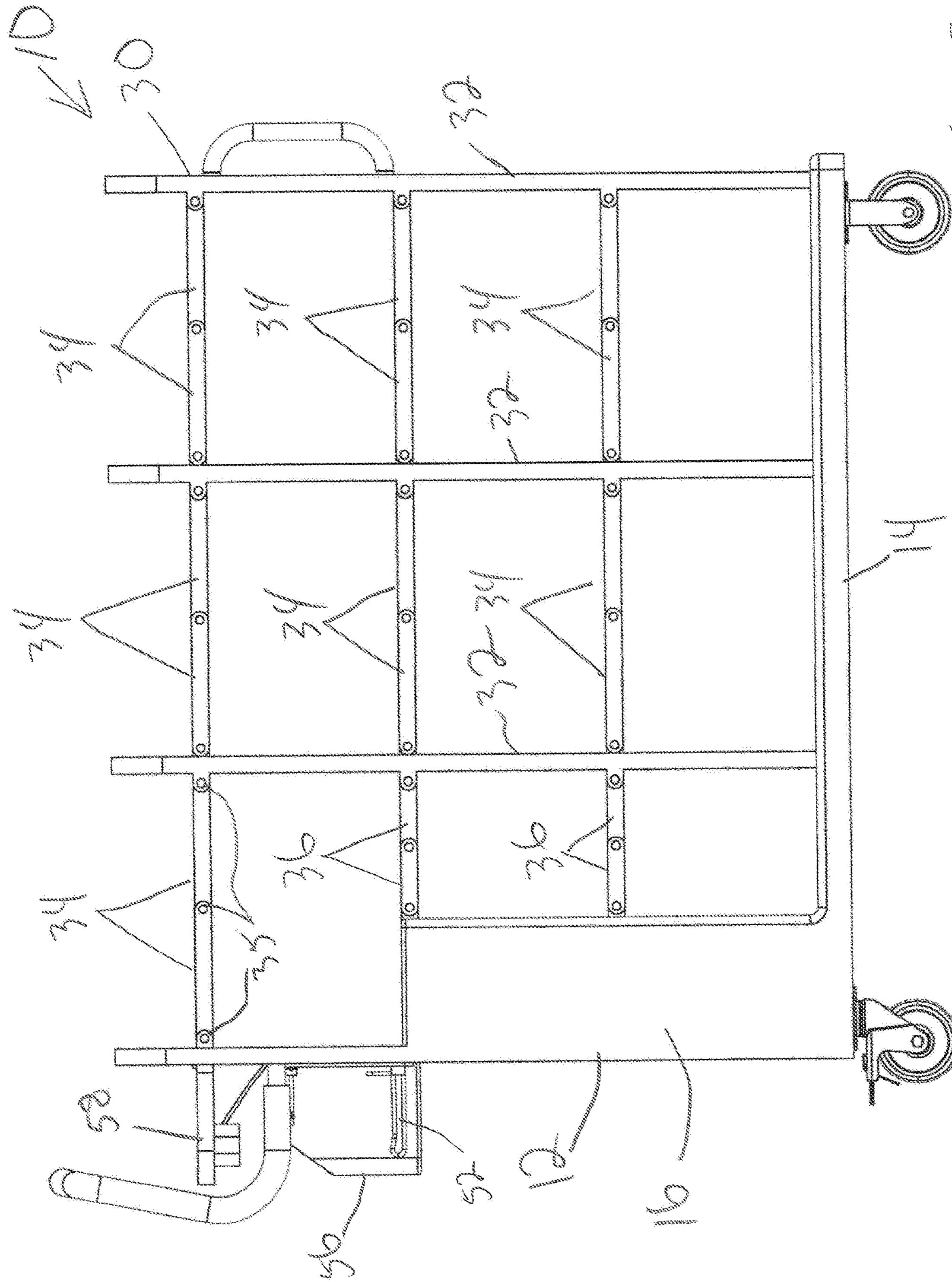


Figure 2



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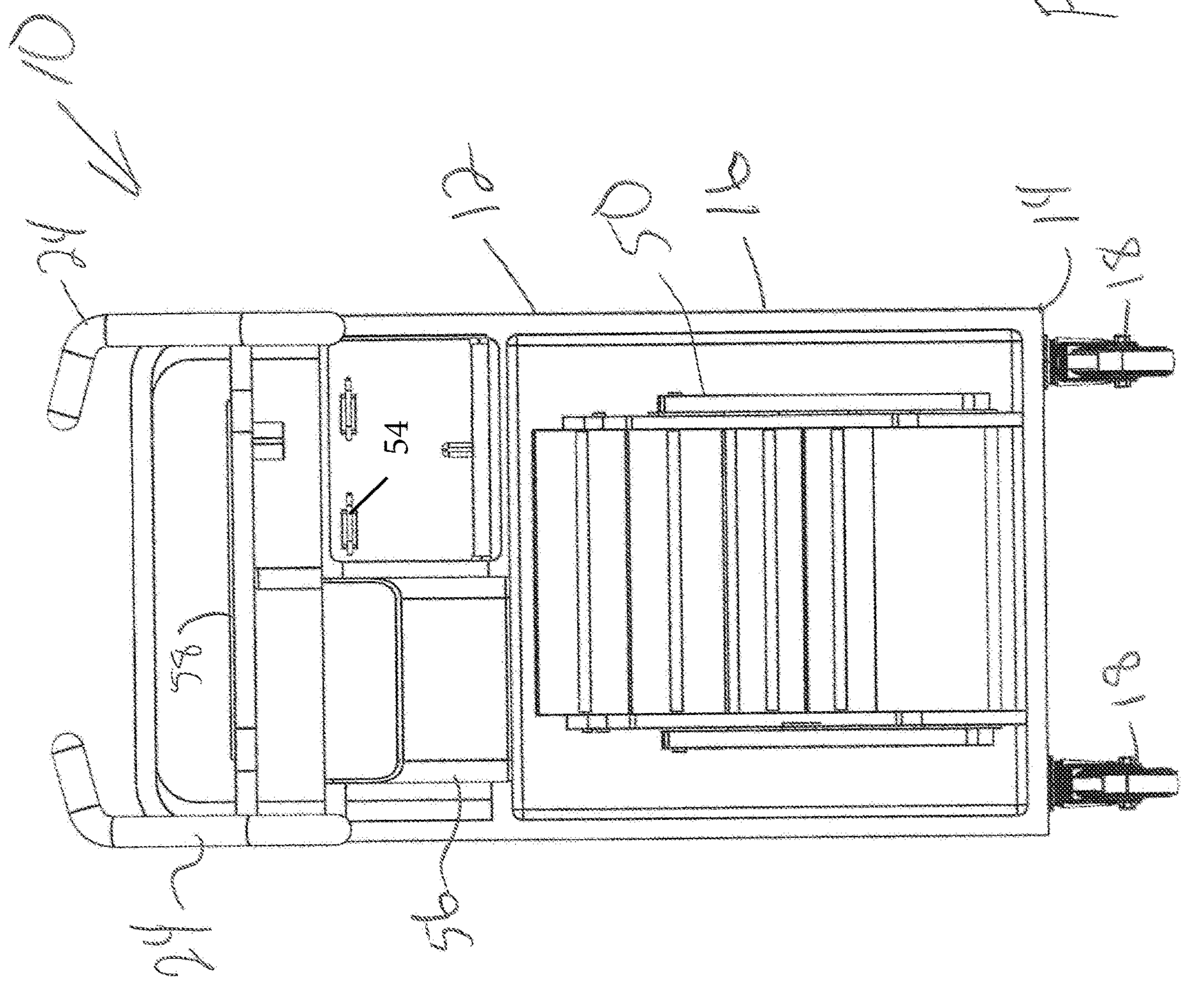

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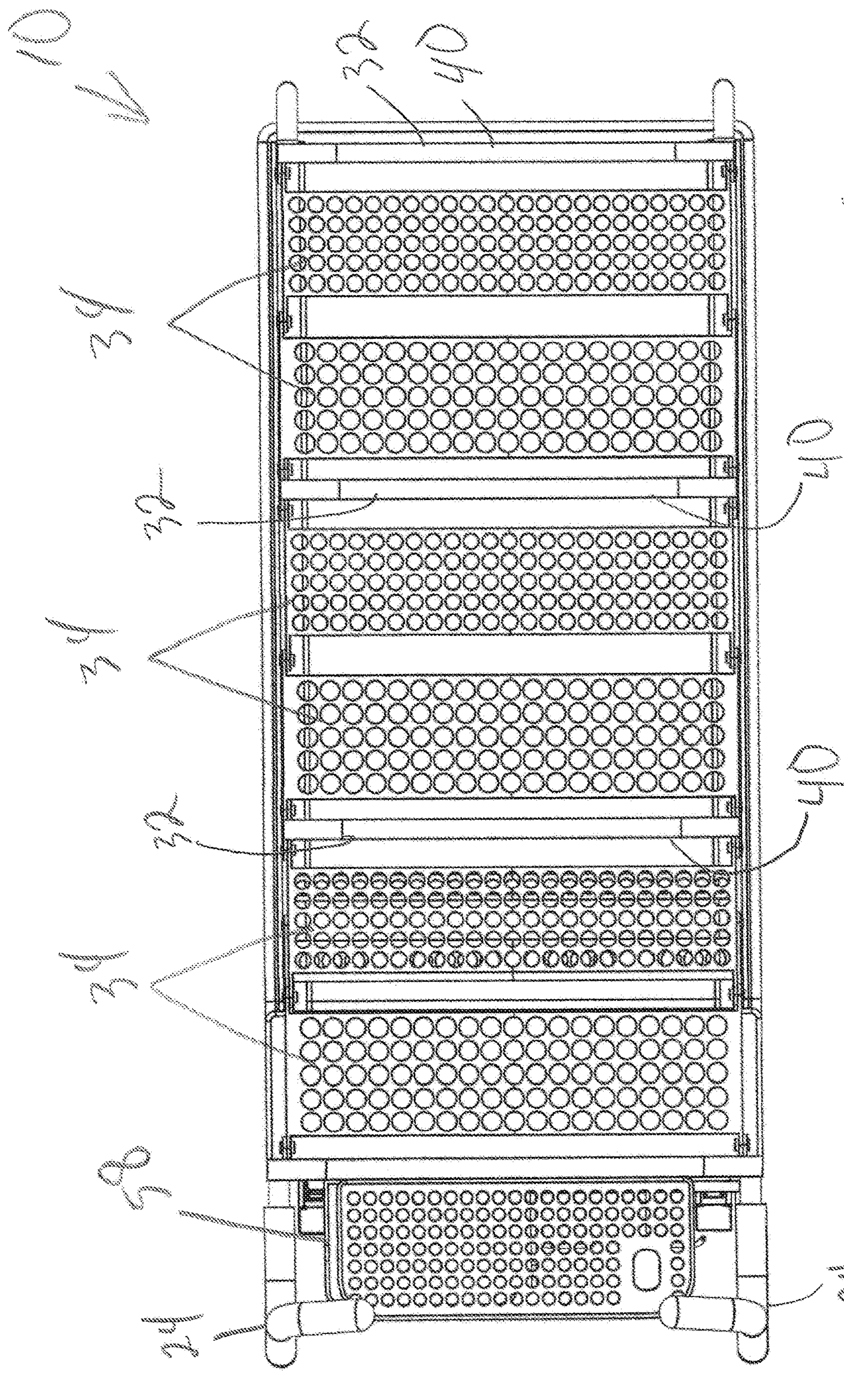

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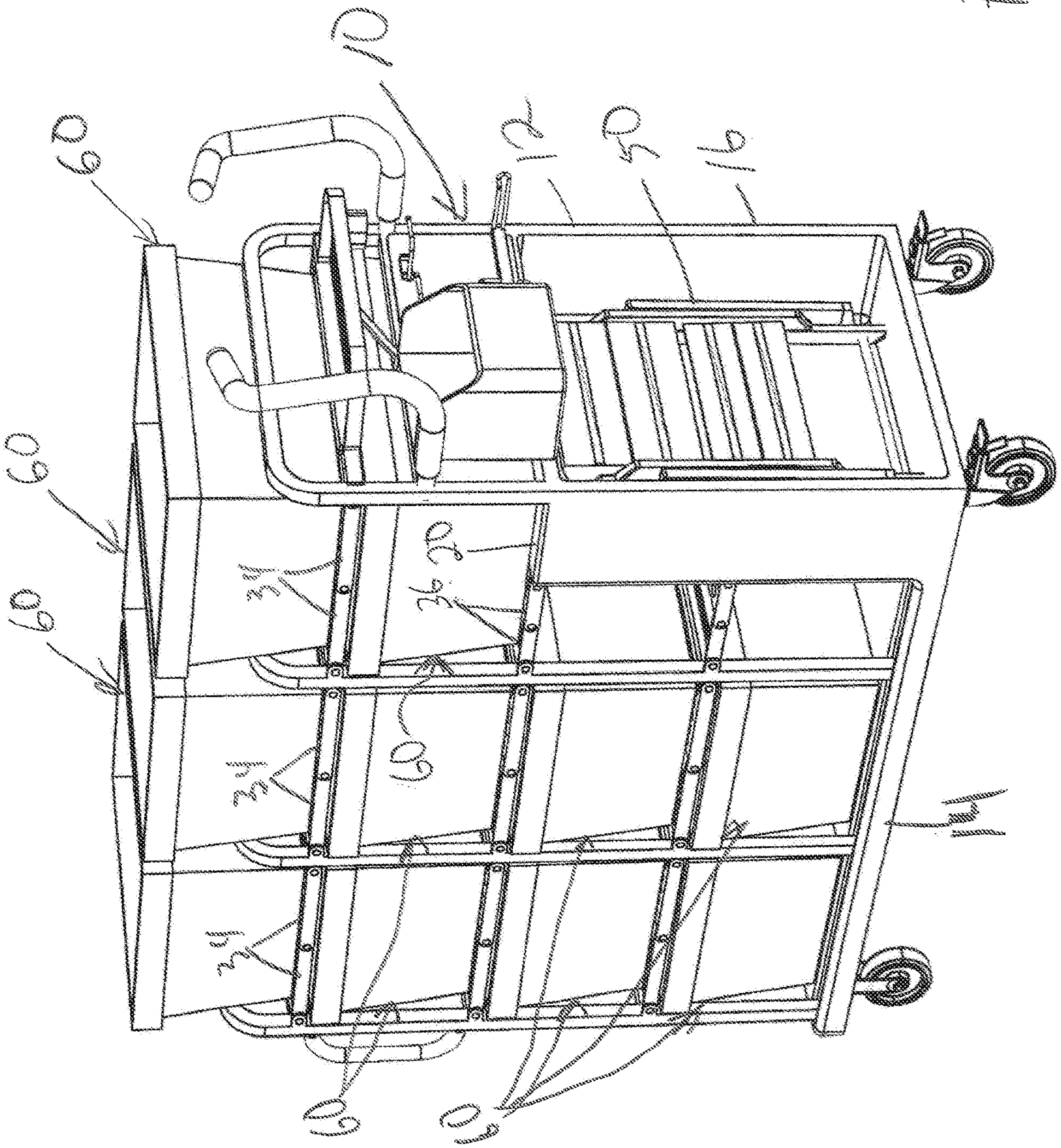

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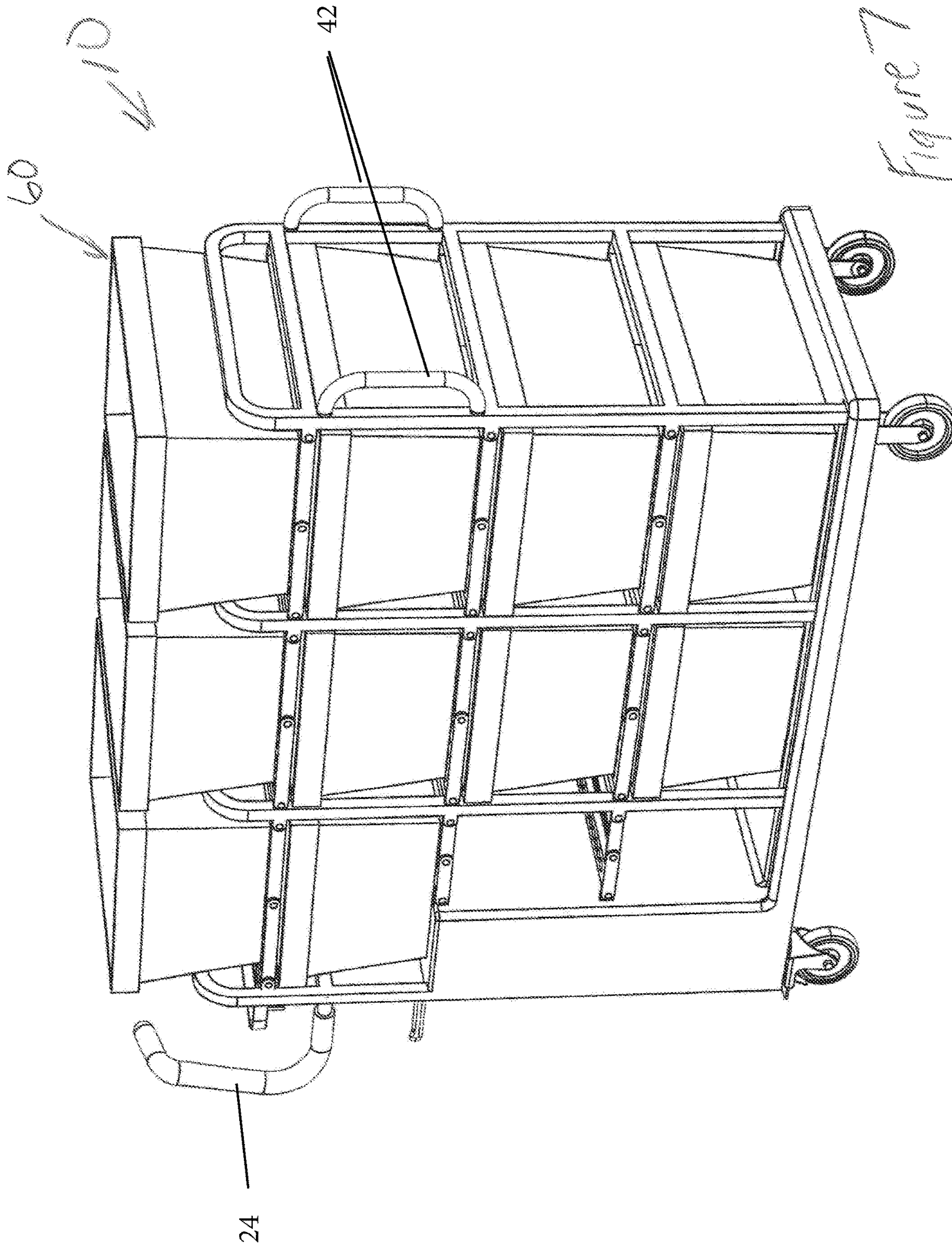

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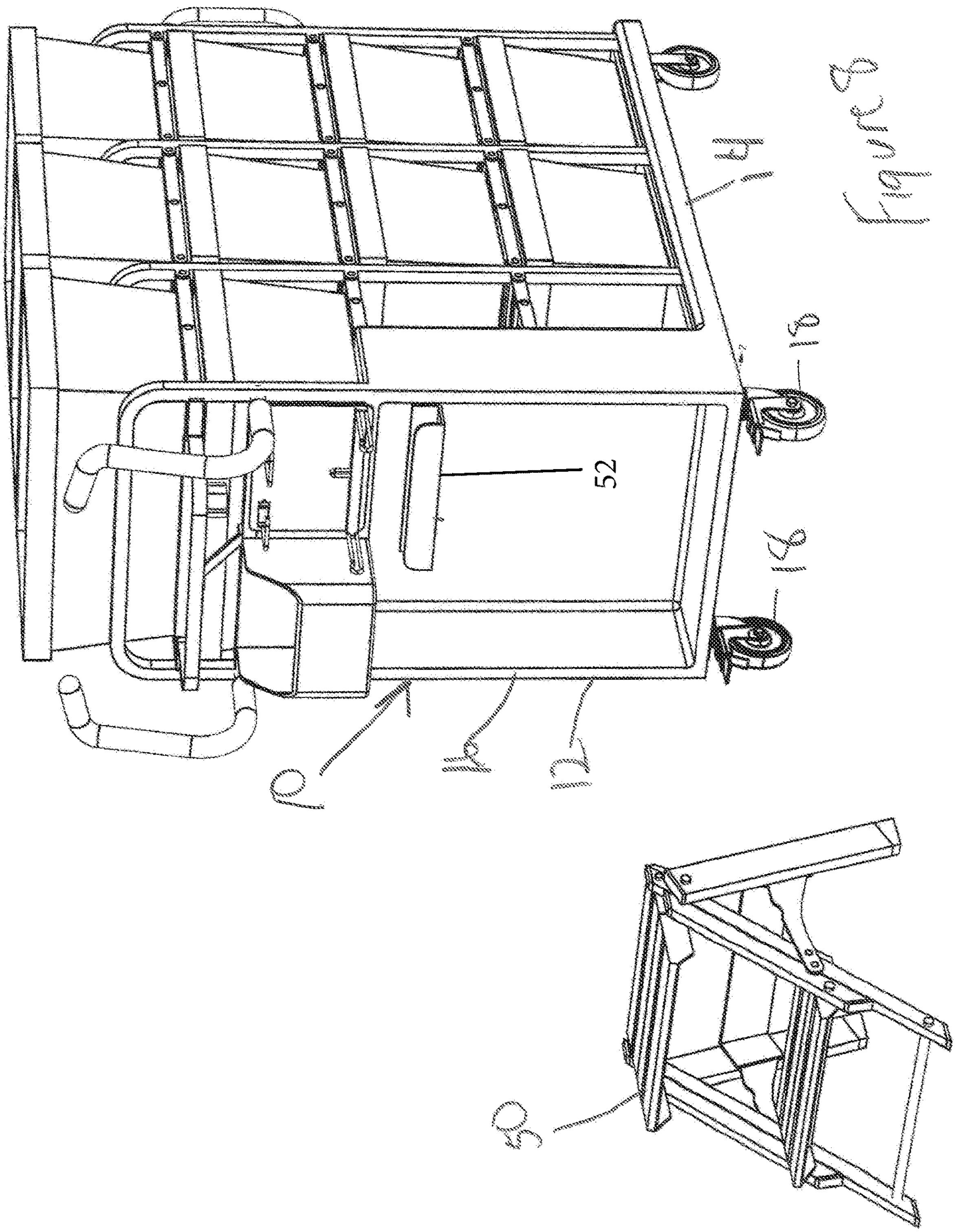

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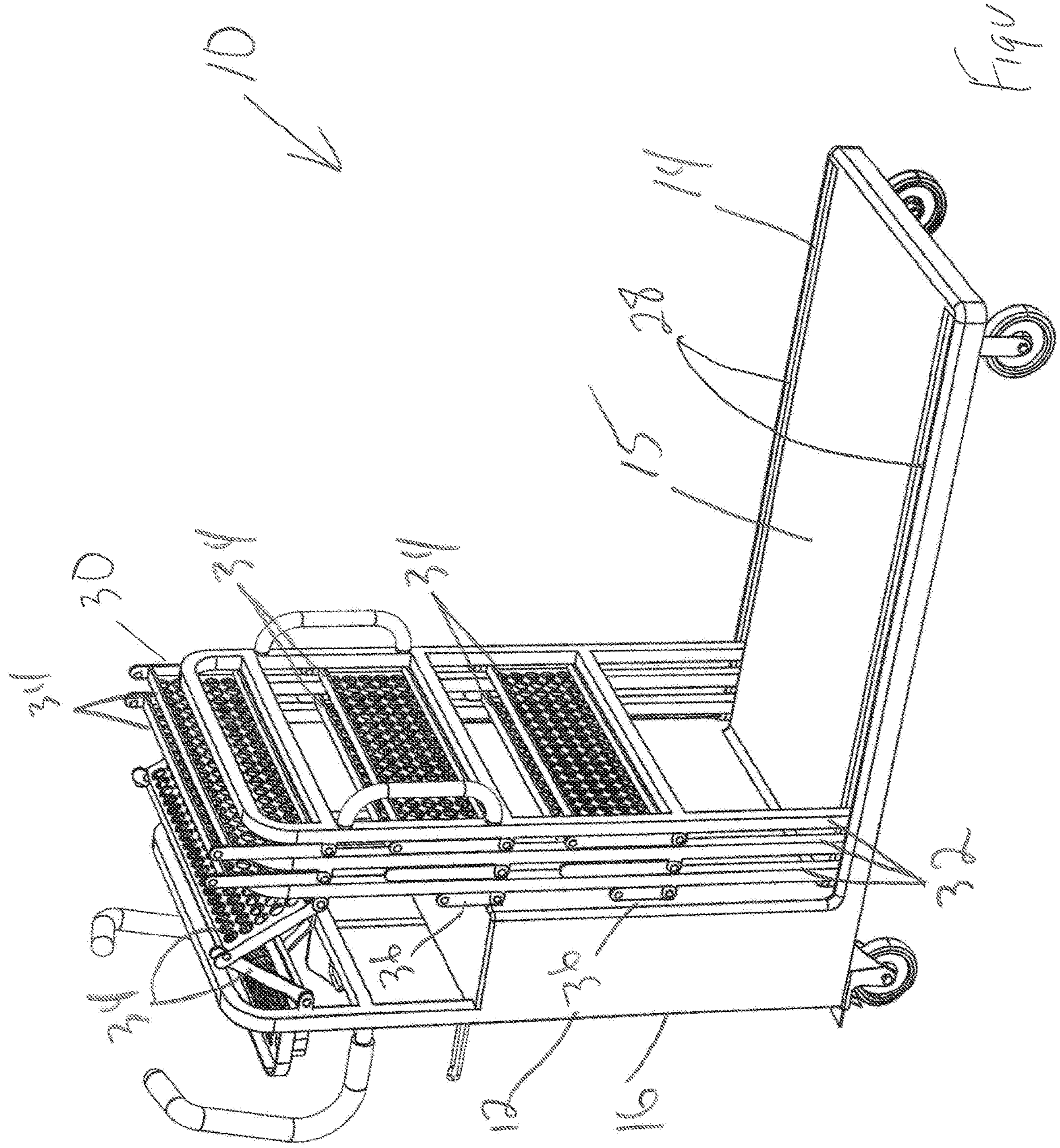

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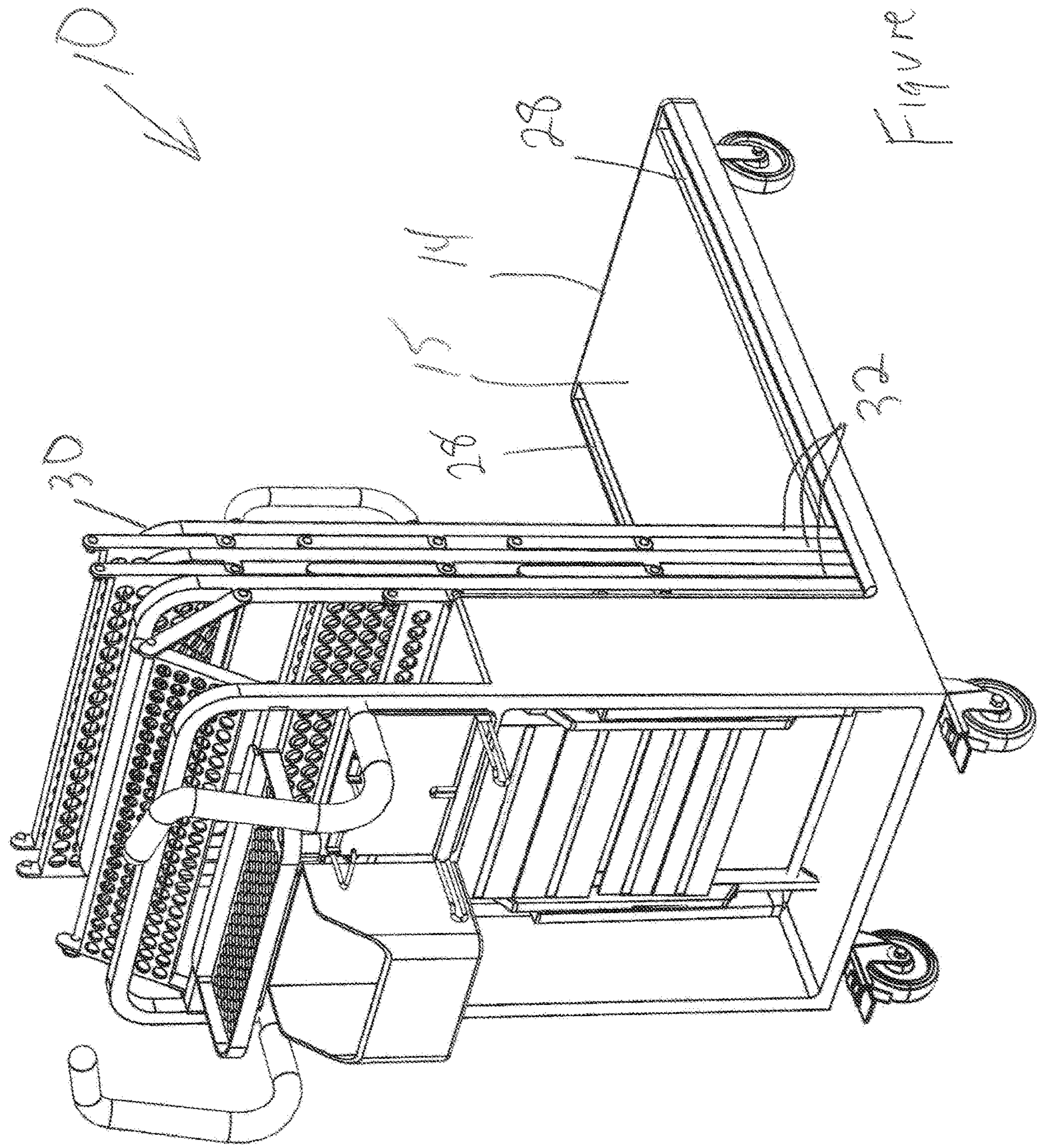

Figure 1D

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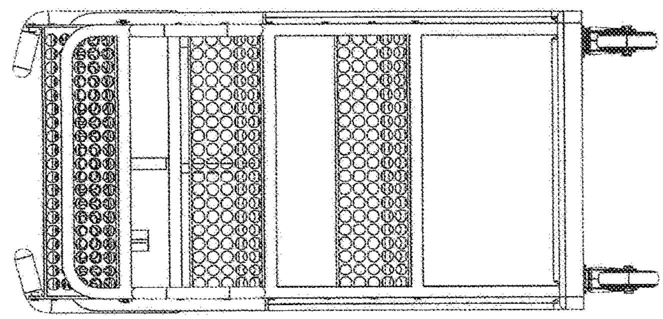
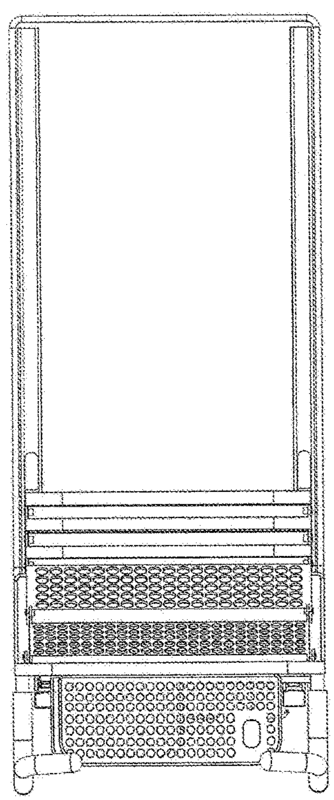


Figure 13

Figure 11



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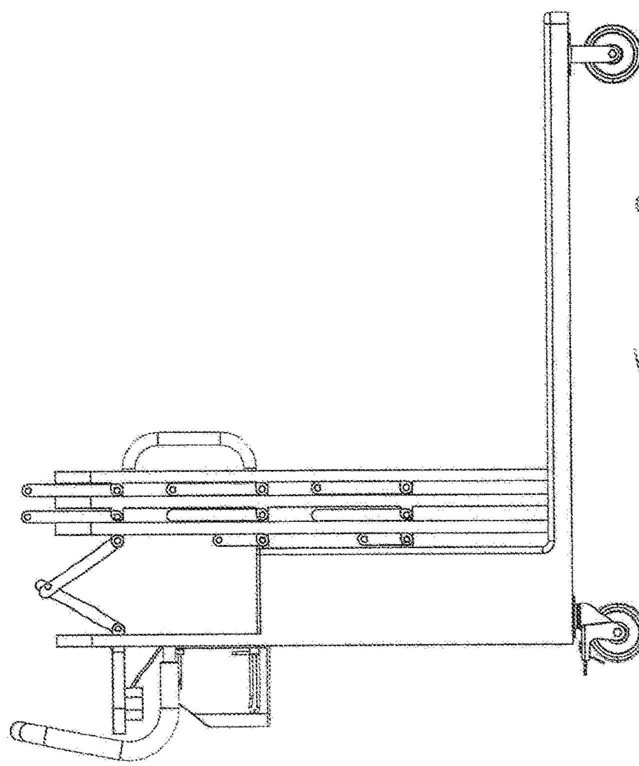

Figure 12

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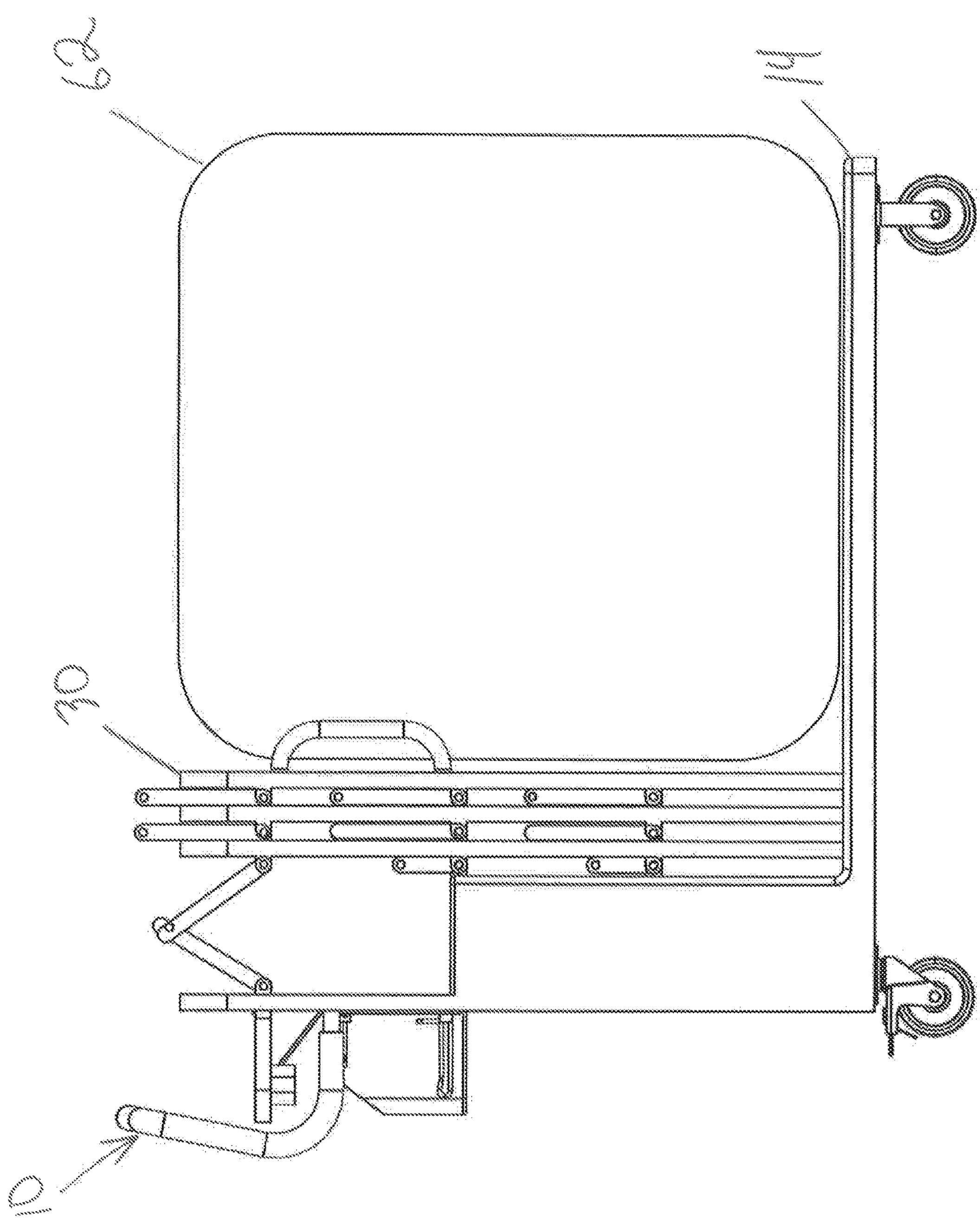

Figure 14

BULK AND TOTE CART

BACKGROUND

It is commonplace for a large format store to have several types of carts available to the store associates for certain processes throughout the store. For example, they may have a bulk cart, commonly referred to as a "L Cart" utilized to transport large items such as TVs. The same store will also have "Tote Carts" utilized for online/app order fulfilment and curbside vehicle delivery. These carts are separate pieces of equipment that are used and stored throughout the location which takes up large amounts of floor space, even when the carts nest.

SUMMARY

In an example embodiment described herein, both functions are combined into a single cart. This gives the store associate order picking flexibility between bulk items or totes so that they do not have to go throughout the store to find the appropriate cart. It also allows the store to consolidate and reduce the total number of equipment onsite along with reducing the lost floorspace.

In one example embodiment presented herein, a cart includes a platform and a plurality of wheels supporting the platform. A plurality of retractable shelves are supported on the platform. The plurality of retractable shelves are reconfigurable between a deployed position and a retracted position. In the deployed position, a plurality of totes (or other smaller objects) can be supported on the shelves. In the retracted position, the cart is in an L Cart configuration. The platform is uncovered by the shelves and a larger item (or several larger items) can be placed on the platform.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** shows the cart **10** in a tote-carrying configuration.

FIG. **2** is a rear perspective view of the cart of FIG. **1**.

FIG. **3** is a side view of the cart of FIG. **1**.

FIG. **4** is a rear view of the cart of FIG. **1**.

FIG. **5** is a top view of the cart of FIG. **1**.

FIG. **6** shows the cart of FIG. **1** with a plurality of totes carried thereon.

FIG. **7** is a front perspective view of the cart and totes of FIG. **6**.

FIG. **8** shows the cart and totes of FIG. **6** with the ladder removed for use.

FIG. **9** shows the cart of FIG. **1** reconfigured to an L-cart configuration.

FIG. **10** is a rear perspective of the cart of FIG. **9** in the L-cart configuration.

FIG. **11** is a top view of the cart of FIG. **9**.

FIG. **12** is a side view of the cart of FIG. **9**.

FIG. **13** is a front view of the cart of FIG. **9**.

FIG. **14** shows a side view of the cart of FIG. **9** with a large item supported on the platform in the L-cart configuration.

DESCRIPTION

A cart **10** according to a first embodiment is shown in FIG. **1**. The cart **10** includes a base **12** having a platform **14** extending forward from a lower end of a housing **16**. The housing **16** extends vertically upward from a rear end of the platform **14**. A plurality of casters **18** support the base **12**. Two of the casters **18** may be fixed and two may be rotatable

about a vertical axis, or all four could be rotatable. Optionally, two or more of the casters **18** could include wheels with hub motors for powering the cart **10**. The housing **16** has an upper surface **20**. A rear frame **22** extends upward from a rear end of the upper surface **20** and a pair of handles **24** extend rearward and upward from the rear frame **22**.

The platform **14** includes a pair of elongated channels **28** extending along long edges thereof, perpendicular to the housing **16**. A retractable shelf system **30** is secured to the housing **16** and received in the elongated channels **28** of the platform **14**. The retractable shelf system **30** is shown in FIG. **1** in the deployed position.

The retractable shelf system **30** includes a plurality (in this example, three) vertical frame members **32** each received in both elongated channels **28**. Between each adjacent pair of vertical frame members **32** are a plurality of shelves (in this example, three) each defined by a plurality of shelf segments **34** (in this example, two). The shelf segments **34** are pivotably connected to one another and to the vertical frame members **32**.

There is also a shelf supported by the rearward-most vertical frame member **32** and the rear frame **22**, which shelf includes a pair of shelf segments **34** pivotably connected to one another and to the vertical frame member **32** and the rear frame **22**.

In this example there are two shelves defined between the rearward-most vertical frame member **32** and the housing **16**, which shelves each includes a pair of smaller shelf segments **36** pivotably connected to one another, to the vertical frame member **32** and to the housing **16**.

Each of the vertical frame members **32** may include a pair of vertical portions **39** having a plurality (e.g. three) of cross-bars **38** connected therebetween and an upper portion **40** connecting upper ends of the pair of vertical portions **39**.

The forward-most vertical frame member **32** may include a pair of handles **42** projecting forward therefrom.

FIG. **1** shows the cart **10** in a tote-carrying configuration, although it could also be used for carrying other smaller items. In this configuration, four levels of support surfaces are created (three levels of shelves plus the platform **14**.

FIG. **2** is a rear perspective view of the cart **10**. The housing **16** opens rearward to define a cavity within which can be carried, for example, a ladder **50**, such as a small stepladder. The ladder **50** can be carried on a hook **52**. Other hooks **54** can be provided to assist with carrying items. A small bin **56** may be mounted within the housing **16**. A rear shelf **58** extends rearward from the rear frame **22**. The rear shelf **58** may be coplanar with the uppermost shelf segments **34**. Optionally, a battery could be provided within the housing for operating hub motors within the wheels of some of the casters **18**.

FIG. **3** is a side view of the cart **10** of FIG. **1**. The shelf segments **34** are connected to one another and to the vertical frame members **32** by hinges **35**.

FIG. **4** is a rear view of the cart **10** again showing the ladder **50**, the hook **54**, and the small bin **56** in the housing **16**. FIG. **5** is a top view of the cart **10**, showing upper support surfaces of the uppermost shelves and shelf segments **34** and rear shelf **58**.

As shown in FIG. **6**, in this configuration, the shelves can be loaded with totes **60**. The totes **60** are supported on each adjacent pair of shelf segments **34** between each adjacent pair of vertical frame members **32**. There is also one tote **60** supported on the upper surface **20** of the housing **16** and the smaller shelf segments **36**. Two totes **60** are supported on the platform **14**. In this configuration, the cart **10** can be used to carry ten totes **60**. None of the totes **60** are supported on other totes **60**, so each can be removed and replaced independently. The totes **60** may be used for picking orders, such as orders place online, app orders, curbside pickup, or the like, where each order is placed into a different tote **60**. In this example, the totes **60** are sized such that they each occupy substantially all of the space in each bay defined by the shelves.

FIG. **7** is a front perspective view of the cart **10** and totes **60** of FIG. **6**. The loaded cart **10** and totes **60** could be pulled or pushed by the handles **42** and/or **24**.

As shown in FIG. **8**, the ladder **50** can be removed from the hook **52** in the housing **16** and after use can be returned to the hook **52**.

FIG. **9** shows the cart **10** reconfigured to an L-cart configuration for moving large items (such as televisions or furniture) on the upper surface **15** of the platform **14**. The retractable shelf system **30** is retracted toward the housing **16** to expose most of the upper surface **15** of the platform **14**. The vertical frame members **32** are slid toward the housing **16**, which causes the shelf segments **34** and smaller shelf segments **36** to pivot upward to a substantially vertical orientation. The cart **10** in the L-configuration can be used to move a large item and then returned to the shelf configuration of FIG. **1** for carrying totes again. The cart **10** can be returned to the L-cart configuration whenever it is necessary to move a large item.

FIG. **10** is a rear perspective of the cart **10** of FIG. **9** in the L-cart configuration. FIG. **11** is a top view of the cart **10** of FIG. **9**. FIG. **12** is a side view of the cart **10** of FIG. **9**. FIG. **13** is a front view of the cart **10** of FIG. **9**.

FIG. **14** shows a side view of the cart **10** of FIG. **9** with a large item **62** (such as a television or other large box) supported on the platform **14** in the L-cart configuration. The large item **62** is supported on the platform **14** in front of the retracted retractable shelf system **30**.

Optionally, the cart **10** may also be provided with a power braking system, which can lock two or all of the wheels when the user presses a button and/or whenever the ladder **50** is removed from the hook **54**. If hub motors are provided in at least some of the wheels, braking can be provided via the hub motors.

In accordance with the provisions of the patent statutes and jurisprudence, exemplary configurations described above are considered to represent a preferred embodiment of the invention. However, it should be noted that the invention can be practiced otherwise than as specifically illustrated and described without departing from its spirit or scope.

What is claimed is:

1. A cart comprising:
- a platform;
- a plurality of wheels supporting the platform; and
- a plurality of retractable shelves supported on the platform, wherein the plurality of retractable shelves are reconfigurable between a deployed position and a retracted position, wherein the plurality of retractable shelves each includes a plurality of shelf segments pivotably secured to one another.

2. The cart of claim **1** wherein the plurality of retractable shelves are supported on a plurality of vertical frame members supported on the platform wherein the plurality of vertical frame members are movable between a deployed position and a retracted position on the platform.

3. The cart of claim **2** wherein the plurality of shelf segments are pivotably secured to the plurality of vertical frame members.

4. The cart of claim **3** further including a housing extending upward from the platform, wherein the plurality of retractable shelves are movable toward the housing to the retracted position and away from the housing to the deployed position.

5. The cart of claim **4** further including a pair of handles mounted to the housing.

6. The cart of claim **4** wherein a first shelf segment of the plurality of shelf segments is pivotably connected to the housing.

7. The cart of claim **6** wherein the housing has an upper surface coplanar with upper surfaces of at least one of the plurality of retractable shelves in the deployed position.

8. The cart of claim **7** further including a stepladder stored on the housing.

9. The cart of claim **1** in combination with a plurality of totes supported on the plurality of retractable shelves in the deployed position.

10. The cart of claim **1** in combination with an item supported on the platform while the plurality of retractable shelves are in the retracted position.

11. A cart comprising:
- a platform;
- a plurality of wheels supporting the platform;
- a housing extending upward from the platform;
- a plurality of vertical frame members supported on the platform and movable toward the housing to a retracted position and away from the housing to a deployed position; and
- a plurality of first shelves each supported by an adjacent pair of vertical frame members, wherein each first shelf includes a pair of hingeably connected first shelf segments, each hingeably connected to one of the associated pair of vertical frame members.

12. The cart of claim **11** further including a plurality of second shelves each including a pair of hingeably connected second shelf segments, wherein one of each of the pair of second shelf segments is hingeably connected to one of the plurality of vertical frame members and the other of each of the pair of second shelf segments is hingeably connected to the housing.

13. A method for using a cart including:
- a) deploying a plurality of retractable shelves on a platform, wherein the plurality of retractable shelves are supported by a plurality of vertical frame members including a forward-most vertical frame member and a rear frame member, wherein deploying further includes moving the forward-most vertical frame member away from the rear frame member;
- b) placing a plurality of smaller objects on the plurality of retractable shelves after step a);
- c) retracting the plurality of retractable shelves after step b); and
- d) placing a larger object on the platform after step c).

14. The method of claim **13** wherein step a) includes moving the plurality of vertical frame members so that they are spaced from one another on the platform, a plurality of shelf segments pivotably connected to one another and to the plurality of vertical frame members.

15. The method of claim **13** wherein the larger object is a television or furniture.

16. The cart of claim **2** wherein the plurality of vertical frame members includes three vertical frame members.

17. The cart of claim **1** wherein the plurality of retractable shelves are supported on a plurality of vertical frame members supported on the platform and wherein the plurality of vertical frame members include a rear vertical frame member and a forward-most vertical frame member, wherein the forward-most vertical frame member is movable toward and away from the rear vertical frame member.

18. The cart of claim **17** wherein the plurality of vertical frame members includes three vertical frame members.

19. A cart comprising:

a platform;

a plurality of wheels supporting the platform;

a plurality of vertical frame members supported on the platform, the plurality of vertical frame members including a rear vertical frame member and a forward-most vertical frame member, wherein the forward-most vertical frame member is movable toward and away from the rear vertical frame member; and

a plurality of retractable shelves supported on the plurality of vertical frame members.

20. The cart of claim **19** wherein the plurality of vertical frame members includes three vertical frame members.

* * * * *